

2110211 Introduction to Data Structure

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Lecture 1

a

be

can

Lecture 2

Array & Vector

```
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int main()
{
    vector<int> a = {2,3};
    vector<bool> b = {true,false,true};

    a.push_back(10);
    a.push_back(20);

    a.insert( a.begin(), 99);
    a.insert( a.end(), -5);

    sort(a.begin(),a.end());

    cout << "a is ";
    for (size_t i = 0; i < a.size(); i++) {
        cout << a[i] << " ";
    }
    cout << endl;

    cout << "b is ";
    for (size_t i = 0; i < b.size(); i++) {
        cout << b[i] << " ";
    }
    cout << endl;
}
```

Word Count Problem

```
#include <iostream>
#include <fstream>
#include <string>
#include <vector>
#include <set>
#include <algorithm>
#include "Tokenizer.h"

using namespace std;

void printUniqueWords1(string filename);
void printUniqueWords2(string filename);
```

```

void printUniqueWords3(string filename);
void printUniqueWords4(string filename);
void printWord(string filename);

int main() {
    string filename = "test.txt";
    printWord(filename);
    printUniqueWords1(filename);
    printUniqueWords2(filename);
    printUniqueWords3(filename);
    printUniqueWords4(filename);
    return 0;
}

bool search(string words[], int n, string w) {
    for (int i = 0; i < n; i++) {
        if (words[i] == w) return true;
        n++;
    }
}

void printWord(string filename) {
    int n = 0;

    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        n++;
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

void printUniqueWords1(string filename) { // wastes too much memory
    string words[100000];
    int n = 0;

    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (!search(words,n,token)) words[n++] = token;
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

```

Dynamic Array

```

void printUniqueWords2(string filename) { // expandable array; wastes time
    int cap = 1;
    string *words;
    words = new string[cap];
    int n = 0;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (!search(words,n,token)) {

```

```

        if (n == cap) {
            string *a = new string[2*cap];
            for (int i=0; i<n; i++) a[i] = words[i];
            delete[] words;
            words = a;
            cap *= 2;
        }
        words[n++] = token;
    }
}

tokenizer.close();
cout << "A total of " << n << " words" << endl;
}

```

Word Count by Vector & Set

Vector

```

void printUniqueWords3(string filename) { // slow find time
    vector<string> words;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (words.end() == find(words.begin(), words.end(), token))
            words.push_back(token);
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

```

Set

```

void printUniqueWords3(string filename) { // quick find existence
    set<string> words;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        // if (words.end() == find(words.begin(), words.end(), token)) // slow find
        if (words.end() == word.find(token)) // if can't find then return word.end()
            words.insert(token);
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

```

Lecture 3

std::pair

Basic

```

#include <iostream>

int main() {
    std::pair<int, float> x;
    std::pair<int, float> y;

    x.first = 10;
    x.second = 20.65;
}

```

```

std::cout << x.first << " "; // 10
std::cout << x.second << std::endl; // 20.65

y = x;

std::cout << y.first << std::endl; // 10
std::cout << y.second + 20 << std::cout; // 40.65
}

```

Pair reserves 2 spaces in RAM for first and second elements. When you use `y = x`, it copies values of `x` to `y`. You need to build template by specifying data types in a pair.

Initialize

```

#include <iostream>

int main() {
    // default constructor
    std::pair<std::string, bool> p; // default of that type: {"", false}
    std::cout << "default [" << p.first << " [" << p.second << "]" << std::endl;

    // initialize by { }
    std::pair<std::string, bool> p1 = { "somchai", true };

    // create pair without specifying type by "make_pair"
    std::pair<bool, int> p2;
    p2 = std::make_pair(false, 10);

    std::pair<bool, int> p3(p2); // must be same type p3 = p2

    // more complex pair
    std::pair< std::pair<float, int>, std::string > p4 = { {20.5, -3}, "abc"};
    std::cout << p4.first.first << " " << p4.first.second << " " << p4.second <<
std::endl; // 20.5 -3 abc
}

```

Array inside Pair

```

#include <iostream>

int main() {
    std::pair<std::string*, std::pair<int, float>> p;

    p.first = new std::string[5];
    p.first[0] = "a";
    p.first[1] = "b";
    p.first[2] = "c";

    p.second = std::make_pair(10, 1.5);

    std::cout << p.first[0] << " " << p.second.first << " " << p.second.second;
}

```

Template

- Template allows “same code, different data types”
- `std::pair` is a “class template”
 - In generic term, pair is defined as `pair<T1, T2>`
- To use `std::pair`, we must supply “Type information” to the template what `T1` and `T2` should be.

STD and Namespace

- The “Fullname” of cout is std::cout
- std is a namespace
- We need to use fullname
 - Too lazy? use “using” keywords

```
#include <iostream>
```

```
// this tells C++ that when we say cout, we mean std::cout;
using std::cout;
```

```
int main() {
    // we still need to use std::endl
    cout << "Yes" << std::endl;
}
```

```
#include <iostream>
```

```
// this tells C++ that when we say something that it does not understand, C++ should
try to use std as its namespace
```

```
// this is VERY BAD PRACTICE in real world.
// 10/10 not recommend
// ... but it's ok for this class
using namespace std;
```

```
int main() {
    cout << "Yes" << endl;
}
```

std::vector

A linear storage of a single data type

Basic

```
#include <iostream>
#include <vector>
```

```
using namespace std;
```

```
int main() {
    vector<int> v1;
    cout << "Size of v1 is " << v1.size() << endl; // 0

    vector<int> v2 = {2,3,4};

    cout << v2[1] << endl; // 3
    v1 = v2;
    v1[0] = 20;

    cout << v1[0] << ", " << v1[1] << v1[2] << endl; // 20, 3, 4

    v1.push_back(99);
    cout << v1.size() << endl;
}
```

- Vector starts with empty element

- Use `size()` to get the number of element
- Use `empty()` to check if a vector has any element

Initialize

- With specific size
- With specific size and starting value

```
#include <iostream>
#include <vector>

using namespace std;

void print_vector(vector<float> v) {
    for (size_t i = 0; i < v.size(); i++) {
        cout << v[i] << " ";
    }
    cout << endl;
}

int main() {
    vector<float> v1(10); // size = 10
    print_vector(v1);

    vector<float> v2(5, 3.55); // size = 5, default = 3.55
    print_vector(v2);

    vector<float> v3(v2);
    print_vector(v3);
}
```

Vector Access & Overflow Problem

Access

```
#include <iostream>
#include <vector>

using namespace std;

int main() {
    vector<float> v1(2);
    vector<float> v2(2);

    cout << "-- v1 --" << endl;
    for (int i = 0; i < 7; i++) {
        v1[i] = i; // this automatically assigns value
        cout << i << ": " << v1[i] << endl; // print nonsense if out of range
    }
    cout << "-- v2 --" << endl;
    for (int i = 0; i < 7; i++) {
        cout << i << ": " << v2[i] << endl; // print "4 5 6" due to buffer overflow (take
        // space of v1)
    }
    cout << "using at" << endl;
    v2.at(1) = 99;
    // this will cause exception 'std::out_of_range'
    for (int i = 0; i < 7; i++) {
        cout << ": " << v2.at(i) << endl;
    }
}
```

```

    }
}

```

- Operator [] won't check for 'out-of-range'
 - Reading, writing beyond size is undefined
 - Might crash
 - Grader will give 'x'
- at () will check bound
 - But slower

You should be careful of using spaces, it can access illegal memories and cause weird behaviors.

Modifying Vector

Resizing

- Resize change the size
 - Enlarge will fill with default

```

#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ":";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v(3,10);
    print(v); // 10,10,10,
    v.resize(6);
    print(v); // 10,10,10,0,0,0,
    v.resize(1);
    print(v); // 10,
    v.resize(7);
    print(v); // 10,0,0,0,0,0,0
}

```

Modify

- pop_back
 - Erase last element
- insert(position,value)
- erase(position)
- Careful!
 - Both insert and erase position must be valid
 - If not vald, it can crash

```

#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ":";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

```

```
}
```

```
int main() {  
    vector<int> v(3,8); // v.begin() is called iterator  
    v.insert( v.begin(), 1); // 1, 8, 8, 8  
    v.insert( v.begin()+3, 2); // 1, 8, 8, 2, 8  
    v.insert( v.end(), 3); // 1, 8, 8, 2, 8, 3  
    print(v);  
    v.erase(v.begin()); // 8, 8, 2, 8, 3  
    v.erase(v.begin()+2); // 8, 8, 8, 3  
    print(v);  
    v.pop_back(); // 8, 8, 8  
    print(v);  
}
```

std::find & std::sort

Find

- find(a,b,c)
 - a and b are position (iterator)
 - find c from a to *before* b
 - if not found return b
- Must #include <algorithm>

```
#include <iostream>  
#include <vector>  
#include <algorithm>
```

```
using namespace std;
```

```
int main() {  
    vector<int> v = {9,-1,3,7,5,2,1,4};  
  
    int x = 5;  
    if ( find(v.begin(), v.end(), x) != v.end() ) {  
        cout << "found" << endl;  
    } else {  
        cout << "not found" << endl;  
    }  
  
    if (find(v.begin(), v.begin()+3, 3) != v.begin()+3) cout << "FOUND" << endl;  
  
    // How many "YES" will be printed? (CHEAT QUESTION)  
    if (find(v.begin(), v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES  
    if (find(v.begin(), v.begin()+4, x) != v.end()) cout << "YES" << endl; // YES  
    if (find(v.begin()+4, v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES  
    if (find(v.begin()+4, v.begin()+8, x) != v.end()) cout << "YES" << endl; // YES  
    // First 3 can't even give v.end(), last did find so it gave that position instead  
}
```

Sort

```
#include <iostream>  
#include <vector>  
#include <algorithm>
```

```
using namespace std;
```

```

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ": ";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v = {9,-1,3,7,5,2,1,4};

    print(v); // 9, -1, 3, 7, 5, 2, 1, 4,
    sort(v.begin()+2, v.begin()+6);
    print(v); // 9, -1, 2, 3, 5, 7, 1, 4,
}

```

Functions that Work with Vector

- sort
- find
- min_element
- max_element
- lower_bound
- upper_bound

Need to use *iterator*

Lecture 4

Vector Iterator

Iterator

- Iterator is a *pointer to position*
- First element is `begin()`
- The one *after* the last element is `end()`
- For insert, valid position is from `begin()` to `end()`, inclusive
- For erase, valid position is from `begin()` up to *before* `end()`
- Different vector, different iterator
 - `v1.end()` is not the same as `v2.end()`

Iterator Arithmetic

- We can add by integer to an iterator
- We can subtract two iterators of the same type
- We can use increment (`++`) and decrement (`--`)

```

#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
    vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

    vector<int>::iterator it1 = v1.begin() + 3; // 30
    vector<float>::iterator it2 = v2.end();      // after 2.73

    // getting value at iterator by using "*" operator
    cout << *it1 << endl;      // 30
    cout << *(it2-1) << endl;  // 2.73
    cout << *it1+2 << endl;    // 32
}

```

```

// iterator arithmetics
vector<int>::iterator it3 = it1 + 2;
cout << "data at it3: " << *it3 << endl; // 50
cout << "different of it3,it1: " << (it3 - it1) << endl; // 2

vector<float>::iterator it4 = v2.end();
it4--;
cout << "data of it4: " << *it4 << endl; // 2.73

// this cannot be done
// cout << (it2 - it1) << endl;
}

```

Iterator All Elements by Iterator

```

#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
    vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

    cout << "----v1----" << endl;
    auto it = v1.begin();
    while (it < v1.end()) { // should use != in other iterator types
        cout << it - v1.begin() << ": " << *it << endl;
        it++;
    }

    cout << "----v2---" << endl;
    for (auto it = v2.begin(); it < v2.end(); it++) {
        cout << it - v2.begin() << ": " << *it << endl;
    }
}

```

- We can compare iterator
 - Beware!! Iterator of some data structure does not support > or < comparator
 - But for vectors, it's ok
- We can use auto keyword to automatically define a type of a variable

Iterator Invalidate

Some Functions that Use Iterators

```

#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

int main() {
    vector<int> v1 = {3,0,1,2,4,-3,9,8};
    vector<float> v2 = {10.2, -4, 0.13, 3.14, 2.73};

    auto it1 = min_element(v1.begin(),v1.end());
    auto it2 = max_element(v2.begin()+2,v2.end());

    cout << *it1 << endl; // -3
}

```

```

    cout << *it2 << endl; // 3.14
}

```

- `min_element` and `max_element` get the iterator of min, max element

New Idiom that Iterate All Elements

- Shorter syntax for loop
- Can use reference operator (&)

```

#include <iostream>
#include <vector>
#include <algorithm>
#include <string>
using namespace std;

int main() {
    vector<string> v1 = {"somchai", "somying", "somsak"};

    // range-based for loop
    for (string x : v1) {
        // x is a copy of each element in v1
        cout << x << ", ";
    }

    // using reference
    // x is THE element of v1, meaning we can modify it
    for (auto &x : v1) { x.replace(0,4,"--"); }
    for (auto &x : v1) { cout << x << " "; } // --hai --ing --ak
    cout << endl;
}

```

Iterator Invalidation

```

#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> v1 = {10,20};

    auto = v1.end()-1; // this points to 20
    // we resize (enlarge) v1, now [it] is invalidated
    v1.resize(10);

    // this might not crash but it actually points to somewhere not in v1
    cout << *it << endl;

    // this will crash the program
    // v1.insert(it,99);
    for (auto &x: v1) {cout << x << " ";}
}

```

- Some operation on the data structure *invalidate* existing iterators
 - For vector, this includes *addition* and *deletion* of element (including *resize*)

Set

- Storing distinct data of same type
 - The data type must be *comparable*, i.e., we can tell if a is more or less than b

- Somewhat slow insert
- Fast look up
- Fast erase
- Iterator starts from “minimum element” and goes in increasing value direction
 - Can be used to (somewhat) fast storing

Basic

- Notice that `s` does not include duplicate elements
- Also see that when we iterate, member is sorted
 - This is distinct characteristic of set

```
#include <iostream>
#include <set>
using namespace std;

int main() {
    set<int> s = {4,1,3,2,1,1,3,4};

    cout << "Size of s is " << s.size() << endl; // 4

    s.insert(10);
    s.insert(5);
    s.erase(3);

    cout << "member of s: "; // 1 2 4 5 10
    for (auto it = s.begin(); it != s.end(); it++) {
        cout << *it << " ";
    }
}
```

Somewhat Slow Insert, Iterate but Fast Find

- We will see this in the detail around last part of this course
 - For now, please believe that
 - If there are N elements in the set
 - Insert takes time directly proportional to $\log(N)$
 - Each `it++` or `it--` takes times directly proportional to $\log(N)$
 - Each find takes time directly proportional to $\log(N)$

Set Iterator

Demos Comparing Vector & Set

- Insert
 - Vector is faster than set
- Find random number
 - Set is much faster than vector

Set Iterator

- We cannot do `s.begin() + x`
 - Because, going to the next element (which is the *successor*) in set is not as fast as vector, c++ forbids `begin() + x`
 - alternatively, you can use `auto it = s.begin(); for (int i = 0; i < 0; i++) {it++;}`
 - We cannot *compare by* `>` or `<`
- We can still use `it++` or `it--` to go to the next or previous (*successor* or *predecessor*) or `x`

```
#include <iostream>
#include <set>
using namespace std;
```



```
int main() {
    set< pair<string,int> > s = { {"somchai",5}, {"abc",6}, {"abcd",-3}, {"somchai",-4},
        {"z",0}, {"z",-1}, {"z",9} };

    for (auto &x : s) {
        cout << x.first << "," << x.second << endl;
    }

    cout << "-- find --" << endl;
    auto it = s.find({"z",-1});
    cout << (*it).first << "," << (*it).second << endl;
    it--;
    it--;
    cout << it->first << "," << it->second << endl;
    it++;
    cout << it->first << "," << it->second << endl;
}
```

Additional Function

- set.lower_bound
- set.upper_bound
- set.count

std::map

Associative data structure with same property as set

Map

- Is very similar to Python's dict in usage
- Is internally implemented as a set with "pair" data type
 - *Same properties, same limitations* as set but more convenience to use as associative data structure
- Associative (mapping) between a *Key Type* and a *Mapped Type*

Basic

```
#include <iostream>
#include <map>
using namespace std;

int main() {
    // map between "Key Type" string and "Mapped Type" int
    map<string,int> m;
    m["somchai"] = 10;
    m["somying"] = -5;
    cout << "Size = " << m.size() << endl; // 2

    // accessing unseen key create a map with default value
    cout << "m[\"xxx\"] = " << m["xxx"] << endl; // 0
    // each element is a pair of key type and mapped type
    for (auto it = m.begin(); it != m.end(); it++) {
        cout << it->first << " is mapped to " << it->second << endl;
    } // somchai: 10, somying: -5

    // this will create mapping "abc" to 0 first and then increase it
    m["abc"]++;
    cout << "now size = " << m.size() << endl; // 4
    for (auto &x : m) {
```

```

    cout << x.first << " is mapped to " << x.second << endl;
} // abc: 1, somchai: 10, somying: -5, xxx: 0
}

```

Checking If Map Has This Key?

- Use find()

```

#include <iostream>
#include <map>
using namespace std;

int main() {
    map<int,string> m;
    m[1] = "somchai";
    m[99] = "nattee";

    int k = 99;
    map<int,string>::iterator it;
    if ((it = m.find()) != m.end()) {
        cout << "Key " << it->first << " is mapped to " << it->second << endl;
    } else {
        cout << "Key " << k << " does not exist in m." << endl;
    }

    // this is not the correct way to check if key exists why??
    if (m[k] != "") {
        cout << "exists" << endl;
    } else {
        cout << "does not exists" << endl;
    }
}

```

Requirement of std::set and std::map

- Set data type and map key type must be comparable
 - We must be able to compare *order* of two elements
- Type that we can use directly
 - int, bool, float, string, double, char, ... and most of other numerical data type
 - Pair can also be used if both the types of first and second are comparable
 - Pair compare first then second

Practice Reading C++ Docs

- Both map and set has *insert* and *erase* function
- What is the return value of both function of each data structure?
 - For set<int> s, can we do s.erase(20)
 - For map<string,bool> m, can we do m.erase("Somchai")??
- If we wish to erase element from index 3 to index 4096 in a vector
 - Is there any function from vector that we can easily use?

Pair-Sum Problem

- Pair Sum
 - Given an array of integers, our task is to find whether there exists a pair of elements in the array such that their summation equal to X
- Input:
 - Array of integers (our main array); this is a large array
 - M values of X, for each value X, we have to determine if a pair whose sum equal to X exists.
- Output:

- For each value of X, print “YES” if we found such pair; print “NO” otherwise

Answer

```
#include <iostream>
#include <array>
#include <algorithm>
#include <cmath>

using namespace std;

bool check(int a[], int n, int X);

int main() {
    int a[15] = {323, 232, 138, 230, 437, 176, 127, 111, 329, 1, 296, 12, 3, 2, 6};
    int n = sizeof(a) / sizeof(a[0]);
    int X = 10000;

    cout << check(a,n,X);
}

bool check(int a[], int n, int X) {
    for (int i=0; i<(pow(2,n)); i++) {
        int t = X;
        for (int j=0; j<n; j++) {
            if (i / ((int) pow(2,j+1)) % 2 == 0) {
                t -= a[j];
            }

            if (0 == t) {
                cout << "YES" << endl;
                return true;
            }
            if (0 > t) {break;}
        }
    }
    cout << "NO" << endl;
    return false;
} // My stupid solution where I brute force every possible combinations
```

Lecture 5

Stack

การเพิ่ม/ลบข้อมูลในกองซ้อน

- ข้อมูล เข้าหลัง ออกก่อน (Last_In First-Out)

กองซ้อน : stack

```
bool        empty();
unsigned    size();
T           top();           // ดูบนสุด
void        push(T element); // ใส่ข้อมูล
void        pop();           // เอาออก
```

ข้อควรระวัง

- ใช้ #include <stack>
- top() ตอนไม่มีข้อมูล → พัง

- `pop()` ตอนไม่มีข้อมูล → พัง
- `stack` ไม่มี iterator
 - ดูเฉพาะข้อมูลที่เพิ่งใส่เข้าไปล่าสุด (`top`)
 - ไม่มี `begin()`, `end()` ให้ใช้
 - ถ้าอยากมาดูข้อมูลทุกตัวใน `stack` ต้องจำใจ `pop` ข้อมูลออกมา

ตัวอย่างการใช้งาน Stack

- การตรวจสอบโครงสร้างแบบซ้อนกัน เช่น การใส่วงเล็บ `(){}[]...`
- การจัดเก็บตัวแปรและการทำ `function calls`
- การประมวลผลนิพจน์ทางคณิตศาสตร์ `postFix` → Discrete Math
- การทำ `undo/redo`
- การค้นคำตอบแบบ `depth-first search`
- ...

Parentheses Checking

การตรวจสอบการใส่วงเล็บ

- ถูก : `( { ( ) [ { } ] } )`
- ผิด : เปิดปิดไม่ตรงกัน `( { ] )`
- ผิด : มีปิดมากเกินไป `( { ( ) } ) )`
- ผิด : มีเปิดมากเกินไป `( { ( ) }`
- วิธีทำ
 - อ่านมาทีละตัว
 - ถ้าเป็นวงเล็บเปิด ให้ `push` ลง `stack`
 - ถ้าเป็นวงเล็บปิด ให้ `pop` จาก `stack` มาตรวจสอบว่าเป็นวงเล็บเปิดที่ตรงกันกับวงเล็บปิดหรือไม่
 - เมื่อใดที่อยาก `pop` ถ้า `isEmpty` แสดงว่า ปิดมีมากเกินไป
 - เมื่ออ่านเสร็จหมด `stack` ยังมีข้อมูล แสดงว่า เปิดมีมากเกินไป

Function Call Stack

Using Stack in Java Virtual Machine

- `jvm` uses java stack to store
 - state of method calls
 - parameters and local variables
 - temporary memory for computation (operand stack)

```
public class StackFrame {
    public static void main(String[] args) {
        a(3,2);
        b(5);
    }
    static void a(int x, int y) {
        int z = x/y;
        b(z);
    }
    static void b(int x) {
        ++x;
    }
}
```

We make a stack frame of function calls, each element consists of arguments and returning argument of the function. Function `main` contains `args` and `RA JVM` and calls `a`, `a` will be added to stack and contains `z`, `y`, `x`, and `RA 04:`, then it calls `b`. After `b` is finished, it will come back to line 9 and popped itself. It will do `a` and go to line 4, and so on. Arguments in each functions are separated variables.

```
#include <iostream>
```

```

using namespace std;

void test(int x) {
    if (x > 0) {
        int y = x-1;
        cout << "I am test(" << x << "). continue... " << endl;
        test(y);
        cout << "now x is " << x << endl;
    } else {
        cout << "I am test(0). stop"
    }
}

int main() {
    test(4);
    return 0;
}

```

Output

```

I am test(4). continue...
I am test(3). continue...
I am test(2). continue...
I am test(1). continue...
I am test(0). stop
now x is 1
now x is 2
now x is 3
now x is 4

```

Postfix Evaluation

Infix and Postfix Expressions

- infix (เติมกลาง)
 - $a + b * c / d - 2, (a + b) * c / (d - 2)$
 - use order of operations
 - use parentheses
- postfix (เติมหลัง)
 - $a b c * d / + 2 -, a b + c * d 2 - /$
 - order is from left to right
 - parentheses may be omitted

1	2	+	3	*	4	1	-	/
			3	3	*	4	1	-
					9	4	1	-
					9		3	/
								3

Calculate Postfix Expressions with Stack

- Value of $2 3 + 4 5 - 6 * + ?$
- Use stack to help
- Solution
 - Check each element in postfix from left to right
 - If it is operand, push stack
 - If it is operator, pop elements from stack as many as operator needs, then push the result
 - After complete, answer will be at the top of stack

Convert Infix to Postfix

Use Stack to Convert

- input : infix expression
- output : postfix expression
- Steps
 - Read each element in infix
 - If it is operand, move it to the back of output
 - If it is operator
 - May pop operator to the end of output
 - push operator to stack
 - If all infix is read
 - pop all operators to the end of input

```
string infix2postfix(string &infix) {  
    int n = infix.length();  
    string postfix = "";  
    stack<char> s;  
    for (int i=0; i<n; i++) { // Read each element  
        char token = infix[i];  
        if (token == operand) {  
            postfix += token; // if operand, add to output  
        } else {  
            while ( ! s.empty() && precedence(s.top()) >= precedence(token) ) {  
                postfix += s.top(); // if operator, pop stack to result,  
                s.pop();  
            }  
            s.push(token); // then push new operator  
        }  
    }  
    while (!s.empty()) {postfix += s.top(); s.pop();}  
    return postfix;  
}
```

Priority of Operators

input 2 * 3 + 4

read	output	stack
2		
*	2	
3	2	*
+	2 3	*
4	2 3 *	+
	2 3 * 4	+
	2 3 * 4 +	

(* comes before and is more important than +)

input 2 + 3 * 4

read	output	stack
2		
+	2	
3	2	+
*	2 3	+
	2 3	* +
	2 3 *	+
	2 3 * +	

Code to Compare Priority

```
string infix2postfix(string &infix) {
    int n = infix.length();
    string postfix = "";
    stack<char> s;
    for (int i=0; i<n; i++) {
        char token = infix[i];
        if (priority.find(token) != priority.end()) {
            postfix += token;
        } else {
            int p = priority[token]; // Get priority of the new operator
            while ( !s.empty() && priority[s.top()] >= p ) { // lazy evaluation; left first
                postfix += s.top(); // pop if top operator is not less important than new op
                s.pop();
            }
            s.push(token);
        }
    }
    while (!s.empty()) {postfix += s.top(); s.pop();}
    return postfix;
}
```

Lazy Evaluation

Setting Priority of Operators

- ^ represents power 7
- ^ before + - * /
- * / before + -
- * same as / 5
- + same as - 3

```
map<char,int> priority =
    {{'+',3},{'-',3},{ '*',5},{ '/',5},{ '^',7}};
```

Complications with Parentheses

- In parentheses is like sub-expression
- Open parentheses always push (very high priority)
- Close parentheses is very low priority
- If found, pop until sees open parentheses

```
input    2    *    (    3    +    4    )
```

read	output	stack
2		
*	2	
(	2	*
3	2	(*
+	2 3	(*
4	2 3	+ (*
)	2 3 4	+ (* (pop all)
	2 3 4 + *	

```
int p = priority[token];
while ( !s.empty() && priority[s.top()] >= p ) {
    postfix += s.top();
    s.pop();
}
s.push(token);
```

```

int p = outpriority[token];
while ( !s.empty() && inpriority[s.top()] >= p ) {
    postfix += s.top();
    s.pop();
}
if (token == ')') s.pop(); else s.push(token);

```

Priority of Operators with Parentheses

	+	-	*	/	^	(	)
In expression	3	3	5	5	7	9	1
In stack	3	3	5	5	7	0	

Right to Left Operator

	+	-	*	/	^	(	)
In expression	2	2	4	4	8	9	1
In stack	3	3	5	5	7	0	

Note Power in expression is more important than in stack

input 2 + 3 - 4 ^ 5 ^ 6

read	output	stack
2		
+	2	
3	2	+
-	2 3	+
4	2 3 +	-
^	2 3 + 4	^ -
5	2 3 + 4	^ -
^	2 3 + 4 5	^ -
6	2 3 + 4 5	^ ^ -
	2 3 + 4 5 6	^ ^ -
	2 3 + 4 5 6 ^ ^	-

Conclusion

- Implementations of stack to solve problems
- Main operations: push / pop / top
- Make stack with array
- If reserve enough space, all runs is $\Theta(1)$; constant time

Lecture 6

Queue

- Just like a normal queue in real life.
 - First-in-First-Out data structure
 - One way in (back of queue), one way out (front of queue)
 - push = add data to the back of the queue
 - pop = remove data from the head of the queue

Basic

```

#include <iostream>
#include <queue>
#include <vector>

```



```

using namespace std;

int main() {
    queue<int> q;
    q.push(1);
    q.push(2);
    q.push(3);
    while (q.empty() == false) {
        cout << q.front() << endl;
        q.pop();
    }
    cout << "-- example 2 --" << endl;
    queue<vector<int>> q2;
    vector<int> v1 = {1,2,3};
    vector<int> v2 = {99,88,-1};
    q2.push( v1 );
    q2.push( v2 );
    cout << q2.back()[1] << endl;
    cout << q2.front().size() << endl;
    auto x = q2.front();
    q2.pop();
    cout << x[0] << endl;
}

size_t    q.size()
bool      q.empty()
void      q.push(T data)
void      q.pop()
T         q.front()
T         q.back()

```

Limitation

- Same limitation as stack
 - No iterator
 - No begin(), end()
 - Can only access front and back of the queue
 - If we wish to access all members, we have to pop it all
 - Do not call front(), back(), pop() when the queue is empty

Radix Sort

Queue Application: Fast sorting with no comparison

Overview

- Put all data in an array
- For each digit X , from LSD to MSD
 - PUT TO QUEUE step: Sort by digit X by putting all of data from the array into B queues
 - B is the base of the number
 - For example, for a base 10 number, we will have Queue[0] to Queue[9]
 - Put into the queue labelled with that digit
 - GET FROM QUEUE step: Start from queue 0 to queue $B - 1$, remove data from the queue and put back to the array

Example

Input 115 15 42 305 21 8 403

Round 1 (digit 0), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
					305				
					15				
	21	42	463		115			8	

⇒ 21 42 463 115 15 305 8

Data is now sorted by the last digits, because we pop from queue 0 to 9

Round 2 (digit 1), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
8	15								
305	115	21		42		463			

⇒ 305 8 115 15 21 42 463

Data is now sorted by last two digits, because we goes from queue 0 to 9, which is grouped by digit 1 and the data in each queue is sorted by the last digit.

Round 3 (digit 2), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
42									
21									
15									
8	115		305	463					

⇒ 8 15 21 42 115 305 463

Code

```
#define base 10

int getDigit(int v, int k) {
    // return the kth digit of v (MSD is digit 0)
    int i;
    for (int i=0; i<k; i++) v /= base;
    return v % base;
}

// d = number of digits
void radixSort(vector<int> &data, int d) {
    queue<int> q[base];
    for (int k=0; k<d; k++) {
        for (auto &x : data) {
            q[getDigit(x,k)].push(x);
            for (int i=0; i<base; i++) {
                while(!q[i].empty()) {
                    data[j++] = q[i].front(); q[i].pop();
                }
            }
        }
    }
}
```

Breadth First Search

Queue Application: Gotta generate 'em all

The Problem

- Given a positive integer X
- Start with a number 1, find a sequence of arithmetics operations, either " $\times 3$ ", or " $/ 2$ " that makes 1 into X

- the $/$ 2 is integer division, e.g., $5 / 3 = 1$ (not 1.6666)
- The sequence must be as short as possible
- Example
 - $10 = 1 * 3 * 3 * 3 * 3 / 2 / 2 / 2$
 - $31 = 1 * 3 * 3 * 3 * 3 * 3 / 2 / 2 / 2 / 2 / 2 * 3 * 3 / 2$

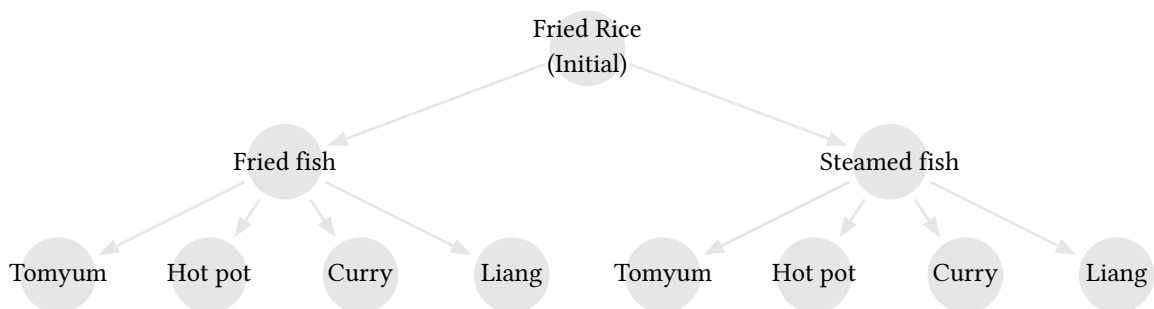
The Idea

- Generate all possible sequences
 - Start with length 1, 2, 3, ... until we find one that gives X
- This is called an *exhaustive search* algorithm
 - Systematically enumerate all possible somethings

Tree Structure (Search Tree)

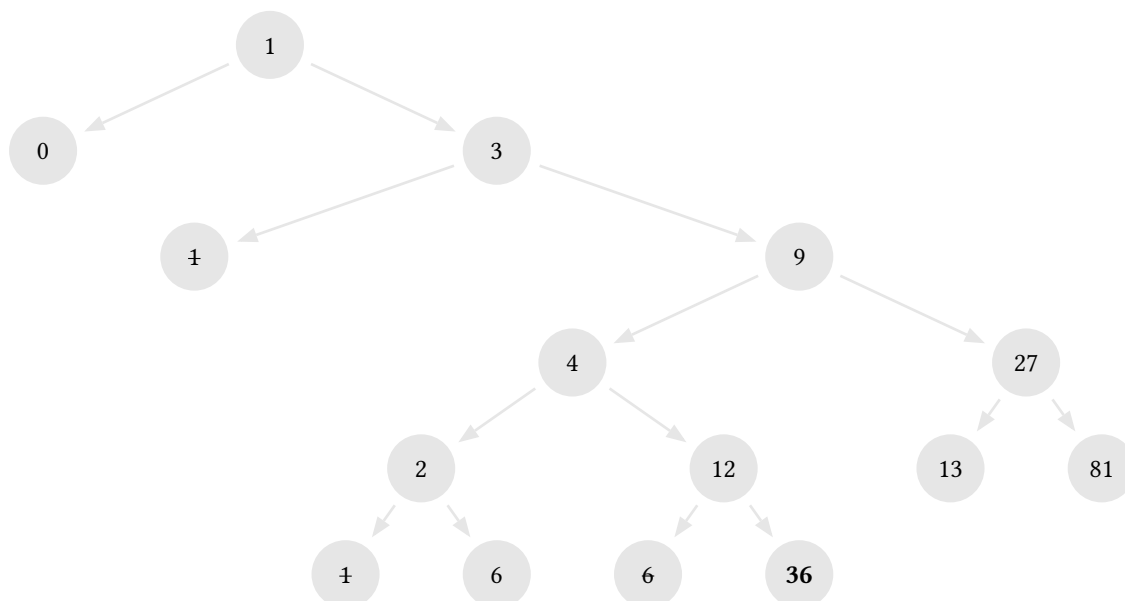
- A structure to illustrate *search* algorithm
- Divide into steps
 - Start with *intial solution*
 - For each possible outcome (called a *state*) of each step, *generate all* proper possible *next step*
 - Also, check if the current step is what we needed
- Written as a diagram of *node* and *edge*

Enumerate



Back to Our Problem

- Start with 1
- Each step is either $* 3$ or $/ 2$
- Issue: might get repeated number
 - Solution: if we have found it, do not generate new step



M3D2 Code

Code

- Queue makes ordering of how we pick a state to enumerate
- From top to bottom and left to right

```
void m3d2(int target) {
    map<int, int> prev;
    queue<int> q;
    int v = 1;
    q.push(1); prev[1] = -1;
    while (!q.empty()) {
        v = q.front(); q.pop();
        if (v == target) break;
        int v2 = v/2;
        int v3 = v*3;
        if (prev[v2] == 0) {q.push(v2); prev[v2] = v;}
        if (prev[v3] == 0) {q.push(v3); prev[v3] = v;}
    }
    if (v == target) showSolution(v, prev);
}
```

Display Solution

x	0	1	2	3	4	6	9	12	13	27	36	39	40
prev[x]	1	-1	4	1	9	2	3	4	27	9	12	13	81

showSolution

```
void showSolution(int v, map<int,int>& prev) {
    string out = "";
    while (prev[v] != -1) {
        if (prev[v] * 3 == v) {
            out = "x3" + out;
        } else {
            out = "/" + out;
        }
        v = prev[v];
    }
    out = "1" + out;
    cout << out << endl;
}
```

$\Rightarrow 36 = 1 * 3 * 3 / 2 * 3 * 3$

Lecture 7

Priority Queue

Queue with privilege

Intro

- Priority Queue is ...
 - A queue with priority
 - Item with high priority is promoted to the front of the queue
 - There is *no back of the queue*
 - Priority is defined by having more value
 - Comparison, by default, is to use operator <, i.e., if item A < B is true, then B *has higher priority*
 - We can have custom comparator

- Has the *same interface* as stack

Example

For intuitive purpose only! This is not really how priority_queue work internally

```
push(10)
< 10      >
```

```
push(30)
< 30 10   >
```

```
push(20)
< 30 20 10 >
```

```
pop()
< 20 10   >
```

```
push(15)
< 20 15 10 >
|
top
```

Basic

```
size_t      q.size()
bool        q.empty()
void        q.push()
void        q.pop()
T           q.top()
* there is no q.back()
```

Limitation

- Same limitation as stack, queue
 - *no iterator*
 - No `begin()`, `end()`
 - Use `#include <queue>`
 - Can only access `top()` of the queue
 - If we wish to access all members, we have to pop it all
 - Do not call `top()`, `pop()` when the queue is empty
- The data type must be *comparable* (similar to set and map)

Class

Quick Summary

- Syntax
 - class declaration must end with ;
 - Function definition can be outside of the class
 - Access modifier is `public:`, `private:`, `protected:`
 - Constructor is a function with the same name of the class with no return type
- Object is a variable (instantiation) of a class
 - When declared, a constructor is called

Example 1

```
#include <iostream>
#include <string>
using namespace std;

class Student {
```

```

public:
    void setFullName(string name, string surname) {
        this->name = name;
        this->surname = surname;
    }
    string getFullName() {
        return "[" + name + " " + surname + "]";
    }
private:
    string name, surname;
};

int main() {
    Student a;
    Student b;
    a.setFullName("nattee", "niparnan");
    cout << a.getFullName() << endl;
    cout << b.getFullName() << endl;
}

```

Declaring class before writing functions

```

class Student {
public:
    void setFullName(string name, string surname);
    string getFullName();
private:
    string name,surname;
};

void Student::setFullName(string name, string surname) {
    this->name = name;
    this->surname = surname;
}

string Student::getFullName() {
    return "[" + name + " " + surname + "]";
}

```

Output

```

[nattee niparnan]
[ ]

```

Example 2: Constructor

```

#include <iostream>
#include <string>
using namespace std;

class Student {
public:
    Student(float score) {gpax = score;}
    void setFullName(string name, string surname) {
        this->name = name;
        this->surname = surname;
    }
    string getFullName() {return "[" + name + " " + surname + "]";}
    bool is1stHonor() {return gpax >= 3.6;}
}

```

```
private:
    string name, surname;
    float gpax;
};

int main() {
    Student a(2.95);
    a.setFullName("nattee", "niparnan");

    cout << a.getFullName() << endl;
    if (a.is1stHonor()) {cout << "YES" endl;} else {cout << "NO" << endl;}
    // Student b; // <-- cannot compile beacuse there is no default constructor
}
```

Operator Overloading

How C++ has a function for each operator

Overview

- Let say we write `a + b` when `a` and `b` is an object of some classes
 - This can be considered the same as calling a function `plus(a,b)`
 - C++ allows us to write a function for many operator and use it as an operator
 - For example we can write a function `times(a,b)` and let it be used as `a * b`
- This is what we called operator overloading

Instead of writing

```
Matrix multiply(Matrix a1, Matrix a2) {
    multiply(a, multiply(b,c))
}
```

We want to write

```
Matrix a, b, c
a * (b * c)
```

Example

```
#include <queue>
#include <iostream>
#include <string>

using namespace std;

// use operator + symbol that will be overloaded
string operator*(string & lhs, const int & rhs) {
    string result = "";
    for (int i=0; i<rhs; i++) {
        result = result + lhs;
    }
    return result;
}

int main() {
    string a = "abc ";
    cout << (a * 3) << endl; // this gives "abc abc abc "
}
```

- Function must be named operator followed by the operator that we will overload
- Some operator takes two parameters (such as `+`, `-`, `*`, `/`, `%`)

- Some takes one (such as ++, --, !, *, &)

Overloading less-than operator

Using with data structure that require sorting

- We have seen several data structure that requires comparability of the data, such as set, map and priority queue
- If we want to use our class with these data structure, we need to tell them how can we use them
- There are multiple ways to achieve this
 - Let us consider operator overloading

Overloading <

- As stated earlier, set, map, and priority_queue use *operator <* to compare two elements
- It does not work if we overload *operator >*

```
class Student {
public:
    Student(float score, string a, string b) {
        name = a;
        surname = b;
        gpax = score;
    }
    bool is1stHonor() {return gpax >= 3.6;}
    // not good, now our data is public
    string name, surname;
    float gpax;
    // overloading <
    bool operator<(const Student & other) const {
        return gpax < other.gpax;
    }
};

int main() {
    Student a(2.95, "nattee", "niparnan");
    Student b(4.00, "attawith", "sudsang");
    cout << (a < b) << endl; // 1
    priority_queue<Student> pq;
    pq.push(a);
    pq.push(b);
    cout << pq.top().name << endl; // attawith
}
```

Custom Comparator

Why custom?

- By overloading *operator <*, we have defined default ordering of that class
- What if we need another ordering, just for this *priority_queue* only
 - For example, *Student* is ordered by *gpax* by default
 - What if we want our *priority_queue* to order by name instead, while keeping the *Student* default ordering elsewhere
 - Better, can we have multiple *priority_queue* with different ordering?
- Can be done via comparator class

Example

```
#include <iostream>
#include <string>
#include <queue>
```



```

using namespace std;

class Student {...};

// Comparator class use () overloading
class StudentByNameComparator {
public:
    bool operator()(const Student& lhs,
                    const Student& rhs) {
        return lhs.name < rhs.name;
    }
};

class GpaxThenName {
public:
    bool operator()(const Student& lhs,
                    const Student& rhs) {
        if (lhs.gpax == rhs.gpax)
            return lhs.name < rhs.name;
        return lhs.gpax < rhs.gpax;
    }
};

int main() {
    Student a(2.95, "nattee", "niparnan");
    Student b(4.00, "attawith", "sudsang");
    Student c(4.00, "vishnu", "kotrajaras");
    cout << (a < b) << endl; // 1
    StudentByNameComparator comp1;
    GpaxThenName comp2;

    cout << comp1(a,b) << endl; // 0; can use like function

    // Use 3 template parameters
    priority_queue<Student, // class to compare
                  vector<Student>, // vector of that class
                  StudentByNameComparator> pq(comp1); // comparator class

    pq.push(a);
    pq.push(b);
    cout << pq.top().name << endl; // nattee

    priority_queue<Student,
                  vector<Student>,
                  GpaxThenName> pq2(comp2);

    pq2.push(a);
    pq2.push(b);
    pq2.push(c);
    cout << pq2.top().name << endl; // vishnu
}

```

Another Method, lambda-function

```

#include <iostream>
#include <string>
#include <queue>
using namespace std;

```

```

int main() {
    auto compare = [](const string& lhs, const string& rhs) {
        return lhs.size() < rhs.size();
    };

    cout << "Result of compare function = " << compare("xxx","z") << endl;

    priority_queue<string,vector<string>,decltype(compare)> pq(compare);
    pq.push("somchai");
    pq.push("z");
    pq.push("abc");
    while (pq.empty() == false) {
        cout << pq.top() << endl;
        pq.pop();
    }
}

```

Output

```

Result of compare function = 0
somchai
abc
z

```

Priority Queue Template

Templating of priority_queue

- priority_queue requires 3 template parameters
- priority_queue<T, Container = vector<T>, Compare = less<T>>
- The first one is required (which is the type of the data)
- The *second* and the *third* is optional (it has default type)
 - *Second* is the container (for now, just don't think about it)
 - *Third* is the class for comparator (the class that we use to compare)
 - This one is default to less<T>

```

#include <iostream>
#include <string>
#include <queue>
using namespace std;

int main() {
    less<int> x;    // comparator that checks a < b
    greater<int> y; // checks a > b

    int a = 10;
    int b = 3;
    cout << x(a,b) << endl; // 0
    cout << y(a,b) << endl; // 1
}

```

Using Comparator for set and map

- To use custom class with set and map, we need to do the same thing, let set and map know how to sort the data
 - Either make default ordering (overload <) in the custom class
 - Or use custom comparator when declare
- For set, the declaration is set<T, Compare = less<T>>
- For map, the declaration is map<Key, T, Compare = less<Key>>

Assignment

- Is any of `vector<int>`, `set<int>`, `map<int, string>`, `queue<bool>`, `stack<vector<int>>` comparable?
 - For any class that is “YES”, how it is ordered?
 - For example, if `vector<int>` is comparable, how `{1, 2, 3}` is compared to `{1, 2, 3, 4}` or `{2, 3, 4}`

Answer

- `vector<int>` compare each element from the left; `{2, 3, 4} > {1, 2, 3, 4} > {1, 2, 3}`
- `set<int>` compare the maximum element () of each set, if equal then look the next max
- `map<int, string>` compare by Key set, then compare max of value set
- `queue<bool>` is not comparable
- `stack<vector<int>>` is not comparable

Summary

Data Structure Summary

Data Structure	Pro	Cons	Remark
<code>pair<T1, T2></code>	Nothings... It just a pair of two data type		
<code>vector<T></code>	• Fast access [] • Fast append $\mathcal{O}(1)$	• Slow find $\mathcal{O}(N)$ • Slow insert, Slow Erase $\mathcal{O}(N)$	
<code>set<T></code>	• Fast find $\mathcal{O}(\log(N))$ • Item is sorted	• Slower to just append data than vector, stack, queue $\mathcal{O}(\log(N))$ • Iterate is also slow • Takes lots of memory $\mathcal{O}(\log(N))$	Require comparator
<code>map<Key, T></code>			• Associative data type • Also require comparator of Key_Type
<code>stack<T></code>	• Very fast push, pop $\mathcal{O}(1)$	• Very limited functionality but has special uses	• No iterator • Order of data coming out depends on something (stack, queue depends on WHEN it is pushed, pq depends on value) • pq requires comparator
<code>queue<T></code>			
<code>priority_queue<T></code>	• Fast get max • Fast delete max • Data is sorted • Memory efficient	• Slower to just append data than vector, stack, queue • Very limited functionality	

More Data Structure

- C++ has more data structure not really covered right now
 - `list` is a vector with faster input / erase but does not have fast access
 - `unordered_set`, `unordered_map` are set and map that the data is not sorted but is much faster
 - `deque` is a queue that can push, pop at both ends
 - `multiset`, `multimap` are set and map that allows duplicated entries

Lecture 8

Create Your Own CP::pair

Intro

- We start writing our own data structure
 - In our own *namespace* (CP)
- We start with *pair*
 - Good introduction on writing a class
 - Some C++ feature we need to use
 - `const`
 - pass-by-value, pass-by-ref
 - Header file

What should we write in a class

- Class name and namespace
- Member *variables* which define what data we want to store in the class
- Member *functions* which define behavior of the class and how member variable is manipulated
 - Include lots of special function such as *constructor*, *destructor*, *overloading*

CP:pair version 0.1 (minimal version)

- Templating
 - Parameter of a code
- No member functions
 - When we declare a pair, T1 and T2 must be declared as a type

```
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;
    };
}

int main() {
    CP::pair<int,string> p1; // In p1, T1 is int, T2 is string
    CP::pair<bool,int> p2;   // In p2, T1 is bool, T2 is int
}
```

Capabilities

- Can hold 2 pieces of data
- Can use template
- Works with operator =
- Has copy constructor
- Cannot check equal
- Cannot check less than

```
#include <iostream>
#include <string>
using namespace std;
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;
    };
}

int main() {
    CP::pair<int,string> p1, p2; // default ctor
    p1.first = 20; p1.second = "somchai";
    CP::pair<int,string> a(p1); // copy ctor
    p2 = p1;

    cout << p2.first << ", " << p2.second << endl;
}

/*
if (p1 == p2) { // won't compile
```

```

    cout << "yes" << endl;
}

if (p1 < a) { // won't compile
    cout << "yes" << endl;
}
*/

```

CP::pair version 0.2 (comparator overload)

```

#include <iostream>
#include <string>
#include <set>
using namespace std;
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;

        // operator
        bool operator==(const pair<T1,T2> &other) {
            return (first == other.first && second == other.second);
        }

        bool operator<(const pair<T1,T2>& other) const { // operator< must be const
            return ((first < other.first) ||
                    (first == other.first && second < other.second));
        }
    };
}

int main() {
    CP::pair<int,string> p1, p2; // default ctor
    p1.first = 20; p1.second = "somchai";
    CP::pair<int,string> a(p1); // copy ctor
    p2 = p1;
    cout << p2.first << ", " << p2.second << endl; // 20, somchai

    if (p1 == p2) {cout << "yes" << endl;} // yes
    if (p1 < a) {cout << "yes" << endl;}
    set<CP::pair<int,int>> s;
    s.insert({1,2});
    // Now we can compare and use less than (and also set, map, priority queue)
    cout << s.begin()->second << endl; // 2
}

```

Const Member

Const Parameter in Function

```

void test(int &x, const int& y) {
    x++; // Okay; doable
    cout << y << endl; // Okay, we only read y
    for (int i=0; i<y; i++) { // Okay, too
        cout << i << endl;
    }
}

```

```
y--; // ERROR: we promised NOT to change y
}
```

- Declared by putting a keyword `const` as a prefix
- `Const` parameter must not be modified inside the function
- Why? So that we know that the function does not modify the data
 - Especially when we pass-by-reference

Difference of Pass-by-reference && Pass-by-value

Pass-by-value

- The *arguments* can be either constants or variables
- Modifying *parameters* (variable inside the function) does not change the *argument's* value
- The *argument's* value is copied to the parameter
 - **SLOWER!!!** Because we have to copy

Pass-by-reference

- The *arguments* must be variables
- The *parameters* are the argument's variables
 - Modify the *parameter* also modify the *argument's* variable
- **Faster**

Const Member Function

```
class ccc {
public:
    int a,b;

    void inspect() const { // This function promises NOT to change anything
        if (a < b) cout << "yes" << endl; // Okay

        // b += 20; // <-- NOT OKAY
    }

    void mutate() { // This function might change something
        if (a < b) a += 10; // Okay
    }
};

void test2(ccc& changeable, const ccc& unchangeable) {
    changeable.inspect();
    changeable.mutate();
    unchangeable.inspect();
    unchangeable.mutate(); // ERROR: attempt to change unchangeable object
}
```

- Declared by putting a keyword `const` after the function declaration
- `Const` member function cannot modify any member data
 - Also cannot call any other function that is not `const`
- Why? So that we know that the function does not modify its data

Constructor & List Initialization

Custom Constructor

- In STL spec of `std::pair`, it has custom constructor called *intialization constructor*

```
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
```

```

    T1 first;
    T2 second;

    // Custom constructor, using initialization list
    pair(const T1 &a, const T2 &b) {
        first = a;
        second = b;
    }
    // When we have a constructor, a default constructor is not auto-generated

    // Operator
    bool operator==(const pair<T1,T2> &other) {...}
    bool operator<(const pair<T1,T2>& other) {...}
};
}

int main() {
    CP::pair<int, bool> p(10,false);
    CP::pair<string, int> q("abc",42), r("",0);

    cout << (q < r) << endl; // 0
    priority_queue<CP::pair<string,int>> pq;
    pq.push(r);
    pq.push(q);
    cout << pq.top().first << endl; // abc

    CP::pair<string, int> x(q);
    CP::pair<string, int> y = x;

    // All below cannot be compiled
    // CP::pair<string, int> w;
    // vector<CP::pair<int, int>> v(10);
}

```

Initialization List

- Instead of writing a code to assign a value to each member, we can use *intialization list*
 - A little bit shorter code
 - Also little bit faster
 - Only way to init *const* member

```

namespace CP {
    template <typename T1, typename T2>
    class pair {
    public:
        T1 first;
        T2 second;

        // custom constructor, using initialization list
        pair(const T1 &a, const T2 &b) : first(a), second(b) { }
    };
}

```

Default Constructor

- A constructor used when we simply declare an object
- Auto-generated by init all members with its default constructor
- Won't be auto-gen if we have any other constructor

```

namespace CP {
    template <typename T1, typename T2>
    class pair {
        T1 first;
        T2 second;

        // Constructor
        pair() : first(), second() { }
        pair(const T1 &a, const T2 &b) : first(a), second(b) { }
    };
}

```

Include & Header file

Include File

- Usually, our data structure (which is written as a class) will be used by several programs in several files.
 - It is better NOT TO copy our data structure code into each each file
- Rather, put our code in a file and include it where it is needed
 - Better if we want to change it
 - Better compilation
- Introducing “.h” files

C++ header file (.h) and #include

- To put a content of one file into another file, we use #include "filename" keyword
- C++ will simply put the content of *filename* into where we #include it
- #include has more benefit
 - Separation of *declaration* (what it is) and *definition* (how it works)
 - Not really explored in this class
- We usually use .h for a file that we will include but that is not a rule, we can use other extension

Problem with include

a.h

```

class a {
    int m1, m2;
};

```

b.h

```

#include "a.h"

```

```

class b {
    a x;
};

```

c.h

```

#include "a.h"

```

```

class c {
    a y;
};

```

main.cpp

```

#include "b.h"
#include "c.h"

```

```

int main() {
    b b1;
}

```

main.cpp (expanded)

```

class a {
    int m1, m2;
};

```

```

class b {
    a x;
};

```

```

// There are multiple copies of class a
class a {
    int m1, m2;
};

```

```

class c {
    a y;
};

```

```

int main() {
    b b1;
    c c1;
}

```



```

    c c1;
}

```

Fix problem

a.h

```

#ifndef A_H
#define A_H
class a {
    int m1, m2;
};
#endif

```

b.h

```

#include "a.h"

class b {
    a x;
};

```

c.h

```

#include "a.h"

class c {
    a y;
};

```

main.cpp (expanded)

```

#ifndef A_H
#define A_H
class a {
    int m1, m2;
};
#endif

class b {
    a x;
};

// class a here is taken out of
// compilation because A_H is defined
#ifndef A_H
// #define A_H
// class a {
//     int m1, m2;
// };
#endif

class c {
    a y;
};

int main() {
    b b1;
    c c1;
}

```

Summary

Major

- **Templating**: allow user to write a code that works with different data type
- **Constructor**: a function called when we create a variable
 - Default constructor
- **Operator overloading**: for less-than and equality
- pass-by-ref vs pass-by-value

Minor

- Header file
- const
- List initialization

Exercise (no grader)

1. Modify operator< so that it compare *second* before *first*
2. Modify operator< so that when we call sort(v.begin(), v.end()) where w is a vector of our pair, it is sorted from *Max* to *Min*
3. Write operator!= and operator>=

Final version CP.h

```

#ifndef _CP_PAIR_INCLUDED_
#define _CP_PAIR_INCLUDED_

```

```

namespace CP {
    template <typename T1, typename T2>
    class pair {
    public:
        T1 first;
        T2 second;

        pair() : first(), second() {}

        pair(const T1 &a, const T2 &b) : first(a), second(b) { }

        bool operator==(const pair<T1,T2> &other) {
            return (first == other.first && second == other.second);
        }

        // 1.
        bool operator<(const pair<T1,T2>& other) const {
            return ((second < other.second) ||
                    (second == other.second && first < other.first));
        }

        // 2.
        bool operator<(const pair<T1,T2>& other) const {
            return ((first > other.first) ||
                    (first == other.first && second > other.second));
        }

        // 3.
        bool operator!=(const pair<T1,T2>& other) const {
            return !(*this == other);
        }

        bool operator>=(const pair<T1,T2>& other) const {
            return !(*this < other);
        }

    };
}
#endif

```

Lecture 9

CP::vector

Our first “real” data structure

Intro

- Now we will create more complex data structure CP::vector
- It can store variable length array
 - Implemented as a dynamic array
- Can be accessed by operator []